This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication.



# Steady State Visual Evoked Potential-based Computer Gaming on a Consumer-grade EEG Device

Nikolay Chumerin*, Nikolay V. Manyakov*, Marijn van Vliet, Arne Robben, Adrien Combaz, and Marc M. Van Hulle, *Senior Member, IEEE*

***Abstract*—We introduce a game in which the player navigates an avatar through a maze by using a brain-computer interface (BCI) that analyzes the steady-state visual evoked potential (SSVEP) responses recorded with electroencephalography (EEG) on the player's scalp. The four command control game, called "The Maze" was specifically designed around a SSVEP BCI and validated in several EEG set-ups when using a traditional electrode cap with relocatable electrodes and a consumer-grade headset with fixed electrodes (Emotiv EPOC). We experimentally derive the parameter values that provide an acceptable trade-off between accuracy of game control and interactivity, and evaluate the control provided by the BCI during gameplay. As a final step in the validation of the game, a population study on a broad audience was conducted with the EPOC headset in a real-world setting. The study revealed that the majority (85%) of the players enjoyed the game despite of its intricate control (mean accuracy 80.37%, mean mission time ratio 0.90). We also discuss what to take into account while designing BCI-based games.**

***Index Terms*—brain-computer interface, games, human-computer interaction, electroencephalography, steady-state visual evoked potentials.**

## I. Introduction

With a brain-computer interface (BCI), a subject's brain activity is recorded and used for enabling the subject to interact with the external world without involving any muscular activity or peripheral nerves. BCIs are now widely regarded as one of the most successful engineering applications of the neurosciences since they are in a position to significantly improve the quality of life of patients suffering from severe motor and/or communicative disabilities (*e.g.*, amyotrophic lateral sclerosis, stroke, brain/spinal cord injury, cerebral palsy, muscular dystrophy, *etc.*) [1], [2]. Apart from clinical applications, researchers have also started to use BCIs in computer games as an *in-lab* testbed for new decoding algorithms and paradigms [3], [4], [5], [6], [7], [8], [9], [10]. In the same time, these BCIs mostly rely on conventional electroencephalography (EEG) equipment, which requires specific skills (*i.e.*, positioning the cap, application of conductive gel to the electrodes, verifying the signal quality *etc.*), making it all too cumbersome and time consuming experience for the average user. Also, much EEG research is performed in a shielded room (Faraday Cage) for obtaining better and cleaner (*i.e.*, with less artifacts) signals. For these reasons, BCI games based on EEG have rarely met the general public. In the community this is actually regarded as a challenge for the future [8], [11].

At the time of writing, several commercial BCI games are available on the market. Systems that received a lot of media attention are the ones based on the NeuroSky[1] device, for instance the "Force trainer". It allows the player to raise and lower a ball by controlling the rotational speed of a fan by the player's "concentration" level[2]. Mind Flex[3], produced by Mattel Inc., which is also based on the NeuroSky device, takes this concept a step further and adds a turning knob with which the player can control the position of the fan. The goal is to guide the ball through an obstacle course. Both games do not offer precise control and have caused consumer criticisms regarding whether the player has any control at all[4]. In addition to this, the mentioned games are quite simple compared to what is considered in BCI games research.

BCI games intended for a broad audience are required to be easy to set up, the EEG recording device should be comfortable to wear, and game to work properly even in noisy, uncontrolled environments, so that they can be used everywhere [8], [11]. While the potential of BCI games for entertainment has been reported before [11], to the best of our knowledge, there has been no attempt to create and validate a BCI game satisfying the aforementioned requirements. Even the standard BCI applications such as, for example, the brain signal based spelling devices [12], have rarely been evaluated on a broad audience [13], [14], [15]. The majority of the existing BCI games do not meet the above mentioned requirements, and by consequence are not expected to fulfill the user's expectations. In this article, we present the design, the evaluation, and the assessment of a BCI game based on an inexpensive, consumer-grade EEG device, the Emotiv EPOC. Our study also enabled us to collect feedback from naïve participants, recruited voluntarily from a broad audience, playing a BCI game for the first time.

In attempt to make the BCI game even more attractive to the general public, we wanted our game to satisfy some additional requirements, such as user comfort and the set-up time of the

*Equally contributed authors.
All authors are with the Laboratorium voor Neuro- en Psychofysiologie, KU Leuven, Campus Gasthuisberg, O&N 2, Herestraat 49, 3000 Leuven, Belgium e-mail: {Nikolay.Chumerin, NikolayV.Manyakov, Marijn.vanVliet, Arne.Robben, Adrien.Combaz, Marc.VanHulle}@med.kuleuven.be

[1]http://www.neurosky.com
[2]http://company.neurosky.com/products/force-trainer
[3]http://www.mindflexgames.com
[4]http://youtu.be/HsmLA9PqTGM

BCI. By the first criterion, we mean the discomfort caused by the preparation of the set-up and by the electrodes (*e.g.*, dry electrodes potentially do not require any gel or liquid, but cause a lot of discomfort). Therefore, we opted for the consumer-grade EPOC headset. The second criterion refers to the choice of the BCI paradigm. The set-up time can be minimized by selecting a paradigm that does not require any calibration/training (neither for the algorithm nor for the user). We opted for the steady-state visual evoked potential (SSVEP), which is recorded from the occipital pole of the skull. It is the response to a repetitive presentation of a visual stimulus (*i.e.*, flickering stimulus). When repetition is at a sufficiently high rate (starting from 6 Hz), the individual EEG responses overlap, leading to a steady state signal resonating at the stimulus rate and its multipliers (harmonics) [16]. With this paradigm it is possible to detect whether a subject is looking at a stimulus flickering at frequency $f_1$ or not, by verifying the saliency of the frequencies $f_1$, $2f_1$, $3f_1, \ldots$ in the spectrum of the recorded EEG signal (see Fig. 1). Similarly, one can detect which stimulus, out of *several* of them (each one flickering at a different frequency $f_1, f_2, \ldots, f_{n_f}$), is attended to by the subject. By linking each flickering stimulus to a particular command, a multi-command frequency-coded SSVEP-based BCI can be implemented,

Since in the spectral domain the EEG amplitude decreases as $1/f$ [17], the higher harmonics become less prominent. Furthermore, the SSVEP is embedded in other on-going brain activity and (recording) noise. Thus, when considering a recording interval that is too small, erroneous detections are quite likely to occur. To overcome this problem, averaging over several recording intervals [18], or recording over longer time intervals [19] are often used for increasing the signal-to-noise ratio (SNR). An efficient frequency-coded SSVEP-based BCI should be able to reliably detect several ($n_f$) frequencies (see Fig. 1), which makes the SSVEP detection problem more complex, calling for an efficient signal processing and decoding algorithm. Thus, the design of a BCI game is expected to involve a good deal of signal processing and machine learning to ensure a proper detection of the brain activity patterns. It is also expected that (a minimal amount of) decoding mistakes could be for the benefit of the game, turning a shortcoming into a challenge [11]. In this paper, we try to accommodate all these points in a BCI game designed for a broad audience with an easy to use and low-cost EEG device operating in a noisy, uncontrolled environment.

The goal of this study is not only to demonstrate a successful application of a SSVEP-based BCI to computer gaming, but also investigate the perception of the proposed BCI game and its control through *in-lab* experiments as well as via a population study. To evaluate the BCI game one has to assess at least two of its major components: the *game control* and the *gameplay*. The players' *post hoc* subjective assessments of the game's free-play mode provide feedback on the overall perception of the game including the gameplay and the quality of the control. To evaluate the latter on its own, the game should be adapted appropriately. To this end, we eliminated the game elements (*e.g.*, the maze randomness, the freedom to navigate through the game environment, *etc.*) that affect the gameplay but not the game control. This way, the contribution of the gameplay can be decreased in the subjective assessments of the game. Therefore, we split the evaluation of our BCI-game into two parts:

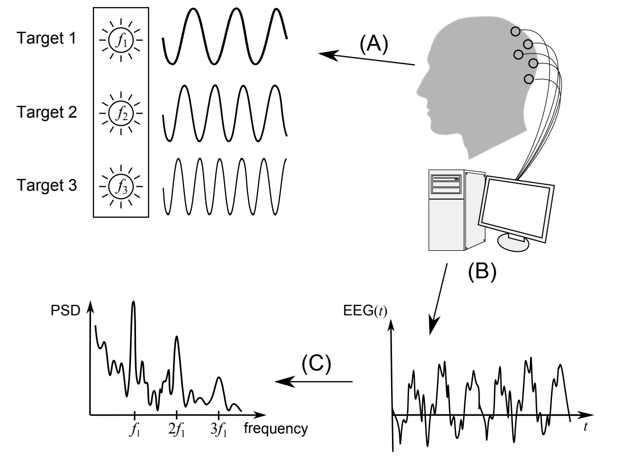



Fig. 1: Schema of the SSVEP decoding approach: (A) subject looks at Target 1, flickering at frequency $f_1$, (B) EEG is recorded and preprocessed, (C) salient peaks at $f_1$, $2f_1$ and $3f_1$ in the EEG spectrum suggest Target 1 as the subject's choice.

1) the assessment of the game control only (see Sec. III-B), in which case the game environment is adapted to enable an objective evaluation of the control parameters when the participants are performing an experimental task, and
2) the overall assessment of the game through the players' subjective evaluation of the unrestricted free-play game mode (see Sec. III-C), in which case the participants are not bound by any specific task.

Since the first part of the assessment could influence the overall perception of the game, we performed those assessments on different groups of subjects.

## II. Methods

### A. EEG Data Acquisition

We have considered two wireless EEG recording devices: one with a setup that is commonly considered for BCI research, thus, for an *in-lab* environment, and an inexpensive, consumer-grade device designed for entertainment purposes.

The first device is a prototype of a wireless EEG system consisting of two parts: an amplifier coupled to a wireless transmitter and a USB stick receiver (Fig. 2a, 2b). The wireless EEG system was developed by IMEC[5] and built around their ultra-low power 8-channel EEG amplifier chip (for technical specifications, see [20]). Each EEG channel is sampled at a 12 bit resolution and at 1000 Hz, which we resampled to 250 Hz as it is not needed for our BCI application. We used an EEG-cap with large filling holes and sockets for active Ag/AgCl electrodes (ActiCap, Brain Products, Fig. 2c). The

[5] http://www.imec.be

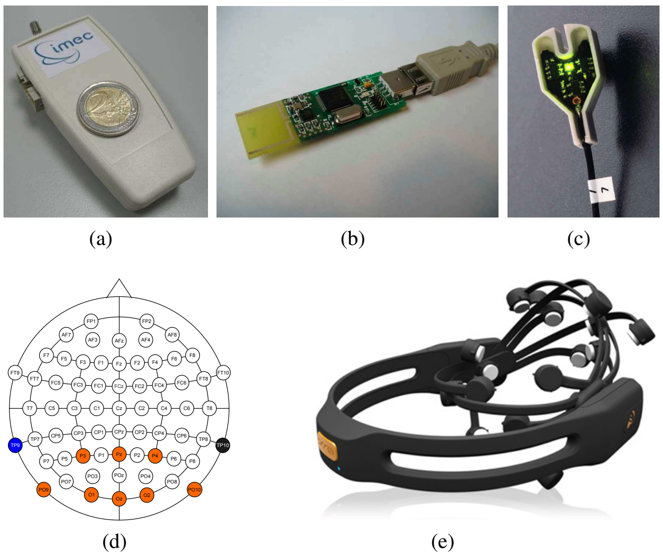



Fig. 2: (a) IMEC wireless eight channels EEG amplifier/transmitter, (b) IMEC receiver, (c) Brain Products ActiCap active electrode, (d) locations of the electrodes on the scalp, (e) Emotiv's EPOC headset.

recordings were made with eight electrodes located primarily on or near the occipital pole, namely at positions P3, Pz, P4, PO9, O1, Oz, O2, PO10, according to the international 10–20 system (Fig. 2d). The reference and ground electrodes were placed on the left and right mastoids, respectively.

The second device is the EPOC (Fig. 2e) headset, developed by *Emotiv*[6]. This headset has 14 saline electrodes and a built-in gyroscope (which is not used in our application). The data are sampled at (reportedly) 128 Hz and resolution of 14 bit/channel/sample and then wirelessly transmitted to a computer. Its low price[7] and wide availability[8] are important features for making BCI games attractive to a broad audience. It is worth to mention that the EPOC was also built as a BCI gaming device, but since it is aimed at exploring inputs from neurofeedback trained EEG patterns, head rotations, and muscular activity (facial expression), it is not directly suited for our purposes as we are aiming at gaming without requiring any training and based only on brain activity. Also, since we are accessing other brain regions than the ones the EPOC was designed for, we had to place the EPOC in a 180°-rotated (in the horizontal plane) position on the head of the player. This way, the electrodes could reach the occipital region (where the SSVEP is most strongly present), instead of the more anterior region for which the device was initially designed. Since the EPOC has a one-size-fits-all design, we cannot precisely describe the electrode locations for a given subject, since they strongly depend on the geometry of the subject's skull. We can only mention the brain area covered by the electrodes. While this could be seen as a drawback from a scientific point of view (not allowing to clearly describe and compare the results between the subjects), it actually increases the usability of the EEG device as one is no longer required to precisely place the electrodes on the scalp, which in turn dramatically reduces the set-up time.

The raw EEG signals from both considered setups were filtered above 3 Hz, with a fourth order zero-phase digital Butterworth filter, so as to remove the DC component and the low frequency drift. A notch filter was also applied to remove the 50 Hz powerline interference.

### B. Visual stimulation

We have used a laptop with a bright 15,4” LCD screen with a 60 Hz refresh rate[9] and, therefore, capable to produce stimulation frequencies up to 30 Hz.

For the lower frequencies, there are at least two options. The first one, which we call *discrete* or *frame-based* stimulation, is commonly used for SSVEP BCIs and normally considers a visual stimulation at frequencies that are dividers of the refresh rate of the screen [21]. An intense (“on”) stimulus is shown for $k$ frames, and a less intense (“off”) one for the next $l$ frames. Hence, the flickering period of the stimulus is $k+l$ frames and the corresponding stimulus frequency is $f_{\text{scr}}/(k+l)$, where $f_{\text{scr}}$ is the refresh rate of the computer screen. For $f_{\text{scr}} = 60$ Hz the stimulation frequencies, thus, could only be 30, 20, 15, 12, 10, 8.57, 7.5, 6.66 and 6 Hz.

In our experiments we used another strategy, which we call *continuous* or *time-based* stimulation. The stimulus' intensity “continuously” changes using some periodic intensity profile $I(t) : \mathbb{R}_+ \to [0, 1]$, (*e.g.*, a sine wave: $I(t) = (1 + \sin t)/2$). Given the stimulus appearance time $t$ in the next video frame, the stimulus should be presented during this frame with intensity $I_f(t) = I(Tft)$, where $T$ is the period of the intensity profile $I(t)$ and $f$ is the stimulus frequency. With this strategy one could, theoretically, present a visual stimulus flickering at any desired frequency $f \leq \frac{1}{2} f_{\text{scr}}$. Using this strategy, one can monitor brain responses even for gradually changing stimulation frequencies (Fig. 3).

### C. Spatial Filtering and Classification

To cancel out nuisance signals as much as possible we use a spatial filter strategy called the minimum energy combination (MEC) method [22]: a linear combination of channels (denoted by a matrix $\mathbf{X}$ where the rows represent channels and samples are in columns) is sought that decreases the noise level of the resulting weighted signals at the frequencies we want to detect (namely, the frequencies corresponding to the periodically flickering stimuli and their harmonics). This can be done in two steps. In the first step, all information related to the frequencies of interest must be eliminated from the recorded signals by using a projection determined by the matrix $\mathbf{P_A} = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$, where the columns of the matrix $A$ consist of discretized values of sine and cosine waves at the *all* frequencies used in the stimulation and their $N_h$ harmonics. The resulting signals $\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{P_A}\mathbf{X}$ contain

[6] http://www.emotiv.com

[7] Starting from $299

[8] More than 30000 devices have already been sold.

[9] The real refresh rate of the laptop was 59.83 Hz, but for the sake of brevity we refer to it as 60 Hz.

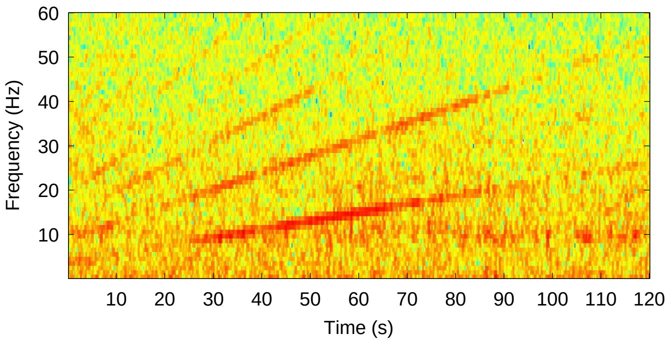



Fig. 3: Spectrogram of an EEG recording for one of the subjects for the continuous stimulation strategy with a gradually changing stimulation frequency (here, the stimulation frequency was linearly changing from 6 to 28 Hz over 120 s). Several oblique lines are visible, corresponding to the stimulation frequencies and their harmonics.

only information that is "uninteresting" in the context of our application, and, therefore, could be considered as noise components of the original signals. In the second step, we look for a linear combination that minimizes the variance of the weighted sum of the "noisy" signals obtained in the first step. This is done through considering the $K$ last (smallest) eigenvalues ($\lambda$) of the covariance matrix $\Sigma = E\{\tilde{\mathbf{X}}\tilde{\mathbf{X}}^T\}$, such that $K$ maximal for $\sum_{k=1}^{K} \lambda_{N+1-k} / \sum_{j=1}^{N} \lambda_j < 0.1$ to be true. The corresponding $K$ eigenvectors, arranged as rows of a matrix $V_K$, specify a linear transformation that efficiently reduces the noise power in the signal $\tilde{\mathbf{X}}$. The same noise-reducing property of $V_K$ is expected to be valid for the original signal $\mathbf{X}$. Assuming that $V_K$ would reduce the variance of the noise more than the variance of the signal of interest, the signal that is spatially filtered in this way, $\mathbf{S} = V_K \mathbf{X}$, would have a greater (or, at least, not smaller) signal-to-noise ratio (SNR) [22]. It is estimated for all stimulation frequencies as

$$Q(f) = \frac{1}{K(N_h + 1)} \sum_{k=1}^{K} \sum_{h=1}^{N_h+1} P_k(hf)/\sigma_k(hf),$$

where $K$ is the dimensionality of the signal $\mathbf{S}$. The signal power function $P(f)$ here is defined as follows:

$$P(f) = \left( \sum_t s(t) \sin(2\pi f t) \right)^2 + \left( \sum_t s(t) \cos(2\pi f t) \right)^2,$$

where $s(t)$ is the signal after spatial filtering, and the noise is estimated according to

$$\sigma(f) = \frac{\pi T}{4} \frac{\tilde{\sigma}^2}{|1 - \sum_{j=1}^{p} a_j \exp(-2\pi i j f / F_s)|},$$

where $T$ is the length of the signal, $i = \sqrt{-1}$, $p$ is the order of the regression model applied to the signals $\tilde{\mathbf{S}} = V_K \tilde{\mathbf{X}}$, $a_j$ are the regression coefficients (estimated, for example, with the use of the Yule-Walker equations), $\tilde{\sigma}$ is the residual regression white noise variance, and $F_s$ is the sampling frequency.

The "winner" frequency $f^*$ is defined as the frequency with the largest index $Q(\,\cdot\,)$ among all frequencies of interest:

$$f^* = \underset{f_1,\ldots,f_{n_f}}{\arg\max}\ Q(f).$$

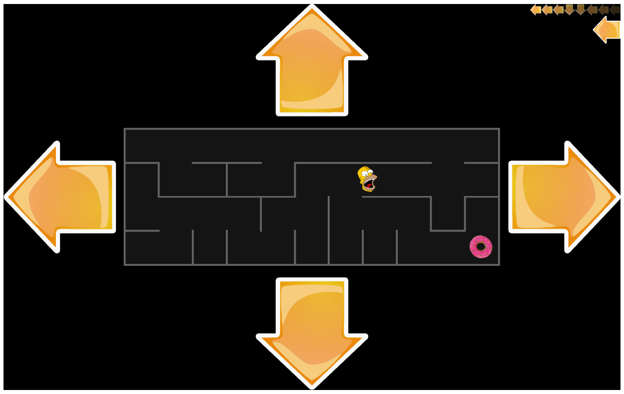
### D. Game Design and Implementation

We have developed an SSVEP-based BCI game called "The Maze", in which the player can control an avatar (the playing character, depicted as Homer Simpson's head) in a simple maze-like environment. The task is to navigate the avatar to the target (*i.e.*, a donut) through the maze (see Fig. 4). The maze consists of square cells (as visualized in Fig. 6) surrounded by walls through which the avatar can not pass. The game has two modes. The default mode is the *free-play* mode (see Fig. 4a), where each maze is generated randomly and the avatar's movement is restricted only by the maze walls. Advancing from level to level, the size of a maze cell decreases allowing to fit more complicated mazes into the same (fixed-size) screen region designated for the maze. For testing purposes we use a special *experimental* game mode with a predefined maze, where the avatar is allowed to move in correct directions only (see Fig. 4b, and Sec. III-B for details).

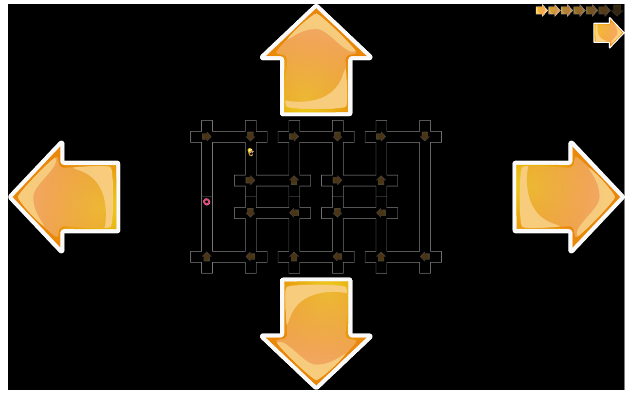
The relative speed of the avatar, defined as the number of cells the avatar can visit per second, is fixed in the beginning of the game and does not depend on the level of the game. We advise to keep the avatar's relative speed quite low (around 0.2–0.5 cells/s) to "hide" the latency discussed further. The player can control the avatar by looking at the flickering arrows (showing the direction of the avatar's next move) placed in the periphery of the maze. Each arrow flickers with its own unique frequency. The choice of those frequencies can be predefined (default settings) or set according to the player's preference.

The game is implemented in MATLAB as a client-server application by using the Psychophysics Toolbox extensions [23] for obtaining a temporally-accurate visual stimulation. Each intermediate decision is produced by the server after analyzing the EEG data acquired over the last $T$ seconds (data window). In the game, $T$ is one of the tuning parameters (which is set before the game starts), which controls the game latency. Decreasing $T$ makes the game more responsive, but at the same time renders the interaction less accurate, resulting in wrong navigation decisions. By default, a new portion of the EEG data is collected every 200 ms. The server analyzes the new (updated) data window and detects the dominant frequency using the MEC method described above. The command corresponding to the selected frequency is sent to the client also every 200 ms, thus, the server's update frequency is 5 Hz.

For the final selection of the command to be executed we weight the items in the queue over the last $m$ commands sent by the server. Each entry of the queue has a predefined weight that linearly decreases from $w_{\text{max}}$ (*i.e.*, the most recent item) to $w_{\text{min}}$ (*i.e.*, the oldest item in the queue). The "candidate" for the "final winner" is selected as the command with the largest cumulative weight. The "candidate" becomes the "final winner" if its cumulative weight exceeds an empirically chosen threshold $\theta = \frac{m}{4}(w_{\text{max}} + w_{\text{min}})$, otherwise no decision is made.



(a) Free-play mode.



(b) Experimental mode.

Fig. 4: Screenshots of "The Maze" game in the (a) free-play and (b) experimental modes. The decision queue (of size $m = 8$) is shown in the upper-right corner. The "final" command is depicted just below the queue. Four big arrows on the periphery are the SSVEP stimuli used to control the avatar (see text).

For the example in Fig. 5 (where the queue length $m$ is 10, the maximal weight $w_{\text{max}}$ is 1, the minimal weight $w_{\text{min}}$ is 0.1, the decision threshold $\theta$ is $2.75 = \frac{10}{4}(1+0.1)$, the cumulative weights are: "left": $\mathbf{3.2} = 1+0.9+0.7+0.5+0.1$, "right": 0.3, "up": $1 = 0.8 + 0.2$ and "down": $1 = 0.6 + 0.4$) the "final winner" is the command "left".

Since the game is BCI-controlled, in order to reach a decision about the avatar's next move, the BCI has to collect and decode a few seconds of EEG data. Thus, the decision is always based on the past few seconds, and this is the source of the latency in the game control. As an attempt to "hide" it, we let the avatar change its navigation direction only in particular cells, which we refer to as the *decision points*: as the avatar starts to move, it will not stop until it reaches the next decision point on its way. This allows the player to use this period of "uncontrolled" avatar movement for anticipating the next navigation direction by looking at the appropriate flickering arrow (see Fig. 6). By the time the avatar reaches the next decision point, the EEG data window, which is to be analyzed, would already contain the SSVEP response corresponding to the *new* navigation direction. Optionally, any cell can be marked as a decision point. However, to make use of the anticipation strategy, the decision points must be distributed sparsely in the maze. In our experiments only the following types of cells were marked as decision points: T-junctions, intersections, turnings, and dead-ends.

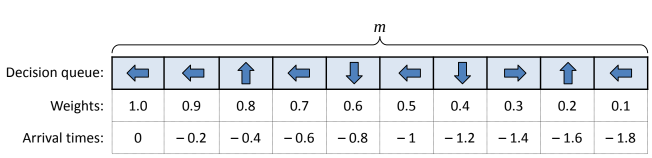



Fig. 5: Example of a decision queue of length $m = 10$ together with the weights of the items in the queue and times of the command arrival (in seconds with respect to the last command arrival).

## E. Evaluation of the game control

We prefer to call the free-play game mode *dynamic*, since the player's behavior depends on the dynamically changing game environment (*i.e.*, the position of the avatar). To assess the efficiency of the dynamic game control we employed two objective measures: first one based on *mission time*, and second based on accuracy. The mission time, in our case, is defined as the time from the beginning of the game level till the moment when the avatar reaches the destination [24]. In order to reduce the intrinsic dependency of the mission time on the maze level complexity, we opted for the *mission time ratio* (MTR), which is the ratio between the *nominal time* and the estimated mission time [24]. By the nominal time we mean the absolute minimal time required to complete the game level, *i.e.*, passing all the decision points without stopping in them (without time penalties)[10].

An MTR that equals 1 indicates perfect control, thus MTR below 1 corresponds to the mission time longer then the nominal time to complete the level. On the other hand, the MTR significantly higher than the one of a random walk suggests that the player *achieved* control over the avatar.

The accuracy measure is defined as the number of correct decisions divided by the number of all decisions taken during a game session. Note, that here (for each decision point)

[10] The time of completion of any game level would be *nominal* if the avatar is controlled by a precise controller (*i.e.*, keyboard) and without player's mistakes.

Fig. 6: The "anticipation" strategy. For the avatar moving from left to right, in order to make the "blue-arrow" turn, the player should start looking at the "down" arrow when the avatar is in blue-shaded cells. The same applies for the "green-arrow" turn and the green-shaded cells. But if the avatar reaches the yellow zone, it would skip both upcoming turns.

we considered only one correct decision out of four possible ones, unlike the accuracy definitions in some other population studies [14], [25].

As it has been discussed in the Introduction, to achieve a fair assessment of the game control, we had to adapt the game by eliminating some game elements, which might affect the perception of the achieved control. The adapted game we implemented in the form of an experimental mode with a special levels (*e.g.*, as depicted in Fig. 4b). All the decision points in this level were placed in "crossings", allowing to leave the decision point in any of four possible directions, only one of which is correct. In order to make the MTR measure more adequate for our purposes, we allowed the avatar to leave a decision point only if a correct decision is made. Otherwise, the decision queue is reset causing the avatar to stay put while the system collects enough data to make a new decision. In this way, the wrong decisions are penalized (by means of time) not so severely as it would be if the avatar was allowed to move in the wrong direction. To complete the experimental level, the participant must pass 20 decision points, eventually making five correct decision in each direction (up, down, left and right). We intentionally designed the experimental level to be as simple as possible, to exclude the necessity of acquiring any advanced playing skills which, for example, might be needed in situations similar to the one depicted in Fig. 6. To simplify the experimental game mode even more, we clearly indicated the correct decisions for each decision point. These markings also allowed subjects to easily discover and understand the decision point mechanism behind the game control. Thus, by removing some challenging elements from the game, we created a controlled game-like environment, where the efficiency of the BCI-based game control can be objectively assessed. In this experiment we also employed the *Visual Analogue Scale* (VAS) [26] for continuous subjective evaluation of the game control between "no control" and "perfect control" extreme levels. The subjective data collected this way were then compared with the objective assessments of the BCI-control of the game.

### F. Overall evaluation of the game

The population study was mostly done to probe the players' feelings toward the SSVEP-based BCI game. Hence, we primary wanted to determine whether this game raises any sense of *fun* for the player. As was shown in [27], fun (namely, hard fun, easy fun, people fun and serious fun) is a key to unlock the players emotion during game play. Thus, for us was interesting to check whether our game can potentially unleash any emotions, not yet going deep into their classification. While we expected that, as a result of the game design, *hard fun* should be detected (playing to see how good I really am; playing to beat the game, requiring a strategy rather than relying on luck, …) [27], we asked about *fun* in general for justifying the game's potential to evoke any emotions, which we consider as one of the important aspect of the game. Since such an assessment could be done only in a free playing mode (where the player does not follow any particular instructions), the assessment was done for subjects other than those participating in the experiment described in Sec. III-B. This *fun* assessment was done to check the attractiveness of the free-play mode of the proposed game.

Additionally, we wanted to check some questions that specially arise from the SSVEP paradigm used for game play. Since SSVEP-BCI uses flicker (to display stimuli associated with the BCI commands), it was reported that it produces visual fatigue, which seems to be less the case with higher frequencies [28], [29]. Since in our game we use low frequencies for controlling, it is quite important to check their usability for BCI gaming. This led to the inclusion of the subject' self-evaluation of visual *fatigue*. On other hand, it is known that the subject's ability to concentrate on the stimuli enhances the SSVEP responses [30], [31]. Thus, for an SSVEP-based BCI game it is particularly interesting to know how easily the player can concentrate on the flickering stimulation during game play. This led to the inclusion of easiness of *concentration* assessment. And, finally, we were interested in the subjects' self-assessment of the game control during game play. Although we already performed a quantitative assessment of control in the previous experiment (see Sec. III-B), which provided us with a real evaluation of the accuracy of our interface, the self-assessment allows us to see how good the gaming elements enabled us to "hide" the not always perfect BCI control during the game.

## III. Experiments and Results

All the experiments were approved by the KU Leuven ethical committee. Prior to the experiments, each participant signed a consent form, among others stating that (s)he had no history of epilepsy[11].

### A. Selection of the game parameters

To run the game, one need to choose its parameters: four stimulation frequencies $f_1, \dots, f_4$, the window size $T$ and the decision queue length $m$. In order to estimate these parameters, and also to investigate the feasibility of using a consumer grade EEG device for the proposed game, we conducted the following experiments.

As for the choice of the stimulation frequencies, we opted for those that are dividers of the screen refresh rate (60 Hz). Such a scheme has already proven itself for SSVEP BCI [21].

We employed the continuous stimulation strategy, which allowed us to reuse the recorded EEG data for further comparison[12] of the continuous *vs.* discrete types of visual stimulation. Following our experience and based on previous findings [32], we selected frequencies 15, 12, 10, 8.57 Hz as the default ones for our game. While other frequencies could also yield good

[11]In photosensitive epilepsy, a seizure could be triggered by flashing or flickering visual stimuli. Most people with photosensitive epilepsy are sensitive to 16–25 Hz (according to Epilepsy Action [http://www.epilepsy.org.uk/]). Up to 5% of people having epilepsy suffer from photosensitive epilepsy (according to the UK Epilepsy Society [http://www.epilepsysociety.org.uk/]). One of the prominent case of mass photosensitive epilepsy was in Japan, where at least 618 children had suffered from convulsions, vomiting, irritated eyes, and other symptoms after watching a particular episode of the "Pokemon" series [http://edition.cnn.com/WORLD/9712/17/video.seizures.update/index.html].

[12]This comparison is not presented here as it is out of the scope of this study.

results, or even better ones, for some subjects, it turned out that the selected ones were satisfactory for all players of the game (which was indirectly supported by our experience both in the *in-lab* and *out-lab* experiments). It is also outside the prominent range of frequencies causing photosensitive epilepsy.

In order to estimate the $m$ and $T$ parameters, we conducted an *in-lab* experiment on six healthy subjects (all male, aged 25–35 with average age 29.2, four righthanded, one lefthanded and one bothhanded). Each subject was presented with a test-level of the game, and was instructed to persistently look at each one of the four flickering arrows for 20 s followed by 10 s of rest. The stimulus to attend to was marked by a small crosshairs symbol. In this experiment the subject's behavior was predefined and did not depend on the game environment, therefore we refer to this experiment as a *static* one. Each recording session consisted of two rounds and, thus, lasted $2 \times 4 \times (20 + 10) = 240$ s. The recorded EEG data was processed off-line using exactly the same methods as during the game. Due to the design of the experiment, the true winner frequency was known at each moment of time. This enabled us to estimate the accuracy of the SSVEP classification.

The game was originally designed and tested on a conventional EEG setup (*i.e.*, IMEC), but for the planned validation of the game, we also tested the game on the EPOC setup. Thus, we had to make sure that the system pre-tuned for the use with the IMEC device *could* also properly worked for the EPOC headset, despite of the "SSVEP-unfriendly" placement of its electrodes. Therefore, we conducted the experiment twice: with the EPOC and the IMEC device. The results of this experiment are presented in Tab. I. By the accuracy of the frequency classification we mean the ratio of the correct decisions with respect to all decisions. Note that the chance level in this case is 25%.

We also conducted a Wilcoxon signed rank test for each value of $T$ and $m$ in order to validate the difference in performance between the two devices. Based on the accuracy results from Tab. I, and supported by our previous experience [33], we choose the window size $T = 3$ and the queue length $m = 5$ (or more) as default values to achieve an acceptable control level. This choice was also motivated by the fact that there is no statistical difference in control accuracy between the considered devices for those parameter values, which justifies the use of a consumer grade EPOC for our game.

We also would like to emphasize that the results in Tab. I should not be considered as any sort of technical comparison of the EEG setups, as this is out of the scope of the present study.

### B. Evaluation of the game control

Twenty healthy, naïve to the game, subjects (15 males and 5 females; aged 15-41 with average age 27.9; 18 righthanded and two lefthanded) participated in the experiment based on the methodology described in Sec. II-E. The tests were performed in an *out-lab* environment very similar to the one the participants would normally experience when playing a computer game. MTR and accuracy estimated from the experimental

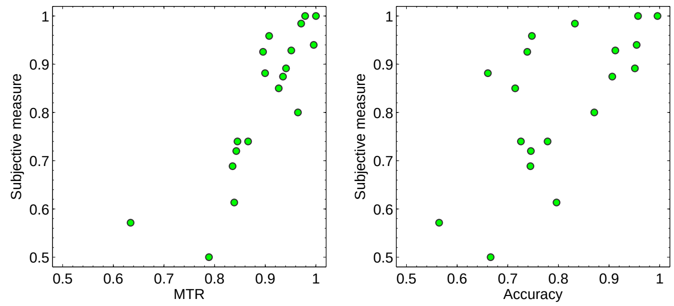



Fig. 7: Results of the game control assessment.

data were $0.90 \pm 0.09$ (mean±std) and $80.37\% \pm 11.85\%$ correspondingly.

We also considered the case of the *random walk* in the experimental mode of the game, when the classifier was generating random decisions (MTR: $0.49 \pm 0.07$, accuracy: $24.68\% \pm 4.61\%$).

The results of the comparison of objective measures *vs.* subjective assessment of game control are shown in Fig. 7. It is clear that the average efficiency of control does not correspond to a perfect control (as it is expected for a BCI system), while it is sufficiently high to consider BCI as a control interface for the proposed game. The existing mistakes could be hidden in a free-play mode (as it was described in Sec. II-D), causing the player to "struggle" even more to "overcome" the hurdles faced when navigating through the maze.

A correlation analysis between objective measures and the subjective assessment of control reveals significant correlation (MTR: $r(18) = 0.86$, $p < 0.0001$, accuracy: $r(18) = 0.65$, $p < 0.0025$), showing us that the subject's perception of the control could also serve as an approximate measure of the interface assessment. The latter is used in the population study in an uncontrolled environment (as the game normally is played) during the assessment of the quality of the gameplay.

### C. Overall evaluation of the game

In order to assess the overall quality of the gameplay, we participated in the *I-Brain & Senses*[13] event. During two consecutive days, 53 persons (aged between 6 and 61 years) played the maze game using several EPOC headsets.

They were mostly high school students during the first day, and persons from all age groups during the second day. The experimental conditions were quite far from those of an *in-lab* environment: a noisy room crammed with all sorts of wireless and other types of equipment (since we shared a room with other demonstration booths). Moreover, the participants were constantly distracted by their companions. These conditions made our game evaluation close to that of a real-world environment where people would typically use such equipment for entertainment purposes (*e.g.*, at their homes or game arcades).

All participants were asked to fill out a questionnaire that consisted of two parts: questions that were asked prior to the

[13]Ghent, Belgium, March 18–19, 2011 http://www.i-brain.be/

TABLE I: Averaged (among 6 subjects) classification accuracy (in percents) for both EEG devices considered as a function of window size $T$ and decision queue length $m$ (see text) and $p$-value of the Wilcoxon signed rank test for the testing the difference between the estimates of the accuracy.

| $T$ | 1 | | | 2 | | | 3 | | | 4 | | | 5 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $m$ | 1 | 5 | 9 | 1 | 5 | 9 | 1 | 5 | 9 | 1 | 5 | 9 | 1 | 5 | 9 |
| EPOC | 59.01 | 59.52 | 60.12 | 82.71 | 83.71 | 84.84 | 91.86 | 92.39 | 92.49 | 96.63 | 96.77 | 96.84 | 98.71 | 98.78 | 98.90 |
| IMEC | 69.60 | 70.38 | 71.21 | 85.80 | 86.86 | 87.57 | 92.24 | 93.03 | 93.31 | 95.90 | 96.43 | 96.75 | 97.05 | 97.68 | 98.04 |
| $p$-value | 0.156 | 0.156 | 0.156 | 0.687 | 0.687 | 0.687 | 1.000 | 1.000 | 1.000 | 0.812 | 0.625 | 0.812 | 0.437 | 0.625 | 0.625 |

experiment and a feedback evaluation after the experiment. In the first part, the participants were asked to indicate their gender (male/female), age, right/left/two-handedness, the number of consumed drinks containing caffeine in the past two hours, the number of cigarettes smoked per day, whether they had any prior experience in EEG experiments, their hair style (thin/thick and bald/long/normal hair). In the second part, we asked each subject to evaluate the required concentration during the game ((1) difficult, (2) neither difficult nor easy, (3) easy to concentrate), visual fatigue ((1) game is tiring, (2) not much, (3) not tiring at all), fun level ((1) the game is fun, (2) not all that fun, (3) unpleasant) and control ((1) good, (2) bad, (3) no control over the avatar).

We did not limit the duration of the gameplay: the participants were allowed to play the game until he/she wanted to stop. Some of the participants succeeded in reaching really complex levels of the game, which indicates their long evolvement in the experiment. Thus, one of the interesting parameters is the subject's perception of being in control of the game. As a result, 36 participants (66%) indicated a good control over the avatar, while 16 participants (30.2%) indicated bad control, and two subjects (3.8%) no control at all (see Tab. II). The latter two (no control) subjects were in a hurry, which was also indicated by them in the questionnaire, thus, they did not spend much time on the game learning to achieve a proper game control. One can see that this subjective evaluation is clearly biased towards being in control. This supports our assumption that the game design special tricks (*e.g.*, described in Sec. II-D), can turn the imperfections of the BCI-based game control into a challenge.

Based on the answers from the questionnaire, we have found that mostly all subjects (45 persons – 85%) had fun with the game, and no one disliked the game (see Tab. II).

In order to properly validate the game, we also asked feedback about fatigue and concentration during gameplay. In spite of the constantly flickering stimuli (which one could regard as irritating), only two participants (3.8%) found them tiring (see Tab. II). But, nevertheless, only 24 participants (45.3%) found them easy to concentrate on (see Tab. II). This means that the game requires some level of concentration on the flickering stimuli, while it does not seem to lead to fatigue, at least not during the time the subject spent on the game.

TABLE II: Results of the overall game evaluation based on the players' self-assessment. Percentage and number of players (between brackets) are shown for each level of the questionnaire (see text for explanation).

| | 1 | 2 | 3 |
|---|---|---|---|
| Concentration | 13.2% (7) | 41.5% (22) | 45.3% (24) |
| Fatigue | 3.8% (2) | 50.9% (27) | 45.3% (24) |
| Fun | 85.0% (45) | 15.0% (8) | 0.0% (0) |
| Control | 66.0% (36) | 30.2% (16) | 3.8% (2) |

We have also analyzed the cross-dependency of different factors. Our results suggests that neither the type of hair, gender, caffeine nor other parameters that were asked prior to the experiment influenced any of the feedback results. A cross-analysis of the feedback reveals that only concentration to the flickering stimuli and self-assessment of the goodness of control depended on each other: easy concentration leads to better control, while difficulties to concentrate causes bad (or lack of) control. The correlation coefficient between those two parameters is $r(51) = 0.70$, $p < 0.0001$, which indicates a significant level of dependency. The results of the subjective assessment are in line with the fact that a proper concentration on the flickering stimuli increases the SSVEP responses [30], which in turn leads to a better detectability of the commands by interface and thus a better control of the game.

## IV. Discussion and Conclusion

### A. General remarks

In this paper, we reported on an SSVEP BCI game that we developed and that was tested on a broad audience. By this study, we tried to bridge the gap between SSVEP BCI games for research purposes, and the testing of such games with a consumer grade EEG device (such as the EPOC). While our results suggest the applicability of a consumer-grade EEG device for BCI gaming, we still see one of the main problem in the one-size-fits-all design of the headset. In our experiments we noticed that for some subjects (young persons) some of the electrodes did not make good contact with the skin, and even sometimes made no contact at all.

Our study could be regarded as an attempt to validate BCI games and to probe their attractiveness on a broad audience, thereby assessing some of the criteria suggested in [11]. The assessment by the players revealed that the BCI game looks attractive and allows for a satisfactory level of control, while requiring some level of concentration. Moreover, the results suggest that the players are able to turn all potential drawbacks (not always perfect classification accuracy, requirement of concentration, necessity to foresee the next command, ...) into a challenge, which contributed to the enjoyment of the gameplay.

The results of the experiment described in Sec. III-B show that for some particular subjects the most/least accurate commands can clearly be seen. While the analysis over all subjects shows no significant difference between the command detection accuracies (repeated measures ANOVA $p = 0.1477$). The erroneous command in the majority of the cases coincided with the previously selected command, which can be explained by the inertia of the game control.

MTR is a continuous measure reflecting the avatar's average speed, which, in turn, can be associated by players with the game control level. Conversely, accuracy is a discrete measure of performance, and arguably less "convenient" for the subjects to estimate and keep track during the game than mission time based MTR. This could explain why the subjective measure better correlates with MTR than accuracy. We included the accuracy analysis as it simplifies the possible comparison of our (game) system with others.

For the chosen game parameters ($T = 3, m = 5$) the accuracy in our static experiment reached $92.39\% \pm 6.96$ (mean$\pm$std) (see Tab. I), which is comparable with other SSVEP BCI systems tested on broad audience 89.16%–95.78% [14], [25]. It is worth mentioning that apart from the fact that we used consumer grade equipment (EPOC) in this experiment, the definition of accuracy we employ is unambiguous and more strict comparing to the ones used in [14], [25].

The accuracy in the dynamic case experiment (see Sec. II-E) reaches $80.37\% \pm 11.85\%$, which in addition to the above mentioned reasons can be explained as follows: a) the subjects had to repeatedly change their attention from the stimuli to the avatar and back, while in the static case the subject could attend exclusively to the stimulus; b) the "inertia" effect caused by the latency of the control and queue-based decision making; and, c) during the gameplay the naïve subject still had to learn the anticipation strategy, which for the static case was not needed.

We believe, these reasons could explain the difference between the static case accuracy reported in Tab. I and the dynamic case accuracy discussed in Sec. II-E.

A few more issues concerning the visual stimulation and the game design need to be discussed. As to the two visual stimulation strategies (introduced in Sec. II-B) it is worth to mention that, at least according to our six *in-lab* subjects, the continuous style (with the sinusoidal intensity profile) is perceptually "easier" on the subject than the discrete one. The continuous style is more robust as it depends only on the next video frame appearance time, which can be estimated quite precisely given the actual video frame appearance time. Even in the case of dropped frames, the continuous/time-based stimulation style is still capable of rendering stable frequencies without phase drifts, which is not the case with the discrete/frame-based strategy. Note also that the continuous style is more general than the discrete one as the latter can be emulated by the continuous style via an appropriate selection of the intensity profile $I(t)$. As a drawback of the continuous style we should mention that, due to the aliasing effect (in the time domain), for some frequencies, the stimuli might be perceived with an undesired intensity amplitude modulation. We defer an in-depth comparison of the two above mentioned visual stimulation strategies to a follow up report.

During the course of our *in-lab* experiments, several subjects mentioned that textured stimuli were easier to concentrate on than uniform ones. Some of our subjects preferred the yellow color of the stimuli over the white one, which partially might be explained by a characteristic feature of yellow light stimulation: it elicits an SSVEP response of a strength that is less dependent on the stimulation frequency than other colors [34]. These clues were considered for the default setting of the game for our broad audience experiments.

While we evaluated through self-assessment some parameters related to the gameplay after one continuous play dependent on player wishes on different subjects (see Sec. III-C), we could assume, that those parameters may change within one subject during the course of gameplay. Thus, for example, long evolvement into the game could lead within same subject to more visual fatigue from stimulation, or concentration could be reduced (leading to less control and, consequently, to the less fun from the game). Thus, it is quite important to investigate the changes of those parameters in time within same subject, to properly assess the BCI gaming for longitudinal use. And this can be seen as a future work.

In this study we only assessed the fun evoked by gameplay in general, so whether it unlocked any emotions [27]. We did not go into the classification of the source of fun, such as a classification into *hard fun* (playing to see how good I really am; playing to beat the game, requiring strategy rather than luck,...), *easy fun* (excitement and adventure, seeing what happens in the story, exploring new world,...), *etc*., leading to emotions such as *fiero* (associated with the moment of personal triumph over one's adversary), *curiosity*, and so on [27]. Or fun can even arise from the fact that one can use his/her brain signals for playing. All this calls for a more in-depth study of which genres of BCI gaming are best suited for evoking emotions by using more rigorous game experience questionnaires and several elaborate BCI games.

### B. SSVEP-based BCI gaming

In this study we constructed a BCI game based on a particular paradigm, namely SSVEP. Since this paradigm requires visual stimulation we, as usual, uses the same screen for the stimulation and the game environment. This is desirable for an SSVEP application to avoid switching between two devices (one for stimulation and one for feedback). Although the SSVEP paradigm is known to provide good accuracy [14] in self-paced static BCI applications without preliminary training, it has its own limitations. Assuming able subjects (not patients with some issues controlling their gaze), we still have to cope with the BCI control latency, the attentional effort, and fatigue caused by the nature of the visual stimulation. As to the possibility to concentrate on flickering stimuli, we found that this could be not always easy and could therefore influence game control. The problem with concentration could arise from the following. The flickering stimulation is somewhat "aggressive" which is why we had to "soften" it by using a continuous stimulation profile (which could lead to a lower accuracy) and/or stimulus color (reducing its luminance contrast), as mentioned in the previous subsection. Another source of concentration problems could be due to the constant shift from the stimuli at the border of the screen to the maze in the center of the screen. As a way to overcome this, we can think about locating the stimuli (arrows) close to the avatar, moving with it during the game or, for example, making the stimuli bigger so that the subject can sit further away from the screen and the avatar is more closer to the fovea when attending the arrows. While such recommendations seem reasonable, they still need to be validated.

Concerning the latency of the SSVEP-based BCI control, we were able to convert it into one of the game's challenges, as described in Sec. II-D. Together with the inaccuracy of the interface, the player could think that some mistakes were made by himself, but not by the BCI. Thus, this motivates the subject to consider them as challenges that need to be met in order to win the game. In this way, we are able to "turning shortcomings into challenges" [11].

Research on BCI-based gaming is still in its infancy, and many more issues need to be addressed before it would become accepted in the gaming community. The issues discussed in this article already indicate the necessity of further research, in general, and the development of suitable applications for interactive entertainment, in particular.

## Acknowledgment

NC is supported by the Tetra project Spellbinder (Flemish Agency for Innovation by Science and Technology), NVM is supported by the research grant GOA 10/019, MvV is supported by IUAP P7/21 AR and AC are supported by IWT doctoral grants, MMVH is supported by PFV/10/008, CREA/07/027, G.0588.09, IUAP P7/21, GOA 10/019 and the Tetra project Spellbinder.

The authors are grateful to Refet Firat Yazicioglu, Tom Torfs and Cris Van Hoof from IMEC Leuven for providing us with the wireless EEG system. The authors are also grateful to Tan Le from *Emotiv* for supplying us some additional EPOC devices for the *I-Brain & Senses* event.

## References

[1] J. Mak and J. Wolpaw, "Clinical applications of brain-computer interfaces: current state and future prospects," *IEEE Reviews in Biomedical Engineering*, vol. 2, pp. 187–199, 2009.

[2] N. V. Manyakov, N. Chumerin, A. Combaz, and M. M. Van Hulle, "Comparison of classification methods for P300 Brain-Computer Interface on disabled subjects," *Computational Intelligence and Neuroscience*, vol. 2011, p. 519868, 2011.

[3] P. Martinez, H. Bakardjian, and A. Cichocki, "Fully online multicommand brain-computer interface with visual neurofeedback using SSVEP paradigm," *Computational Intelligence and Neuroscience*, vol. 2007, p. 94561, 2007.

[4] E. Lalor, S. Kelly, C. Finucane, R. Burke, R. Smith, R. Reilly, and G. McDarby, "Steady-state VEP-based brain-computer interface control in an immersive 3D gaming environment," *EURASIP Journal on Applied Signal Processing*, vol. 19, pp. 3156–3164, 2005.

[5] A. Finke, A. Lenhardt, and H. Ritter, "The mindgame: A P300-based brain-computer interface game," *Neural Networks*, vol. 22, no. 9, pp. 1329–1333, 2009.

[6] J. Bayliss, "Use of the evoked potential P3 component for control in a virtual apartment," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 11, no. 2, pp. 113–116, June 2003.

[7] J. Pineda, D. Silverman, A. Vankov, and J. Hestenes, "Learning to control brain rhythms: making a brain-computer interface possible," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 11, no. 2, pp. 181–184, June 2003.

[8] A. Lécuyer, F. Lotte, R. Reilly, R. Leeb, M. Hirose, and M. Slater, "Brain-computer interfaces, virtual reality, and videogames," *Computer*, vol. 41, no. 10, pp. 66–72, 2008.

[9] R. Scherer, L. Friedrich, B. Allison, M. Pröll, M. Chung, W. Cheung, R. Rao, and C. Neuper, "Non-invasive Brain-computer interfaces: Enhanced gaming and robotic control," in *Proc. IWANN, Part I, LNCS 6691*, 2011, pp. 362–369.

[10] D. Coyle, J. Garcia, A. Satti, and T. McGinnity, "EEG-based continuous control of a game using a 3 channel motor imagery BCI: BCI game ," in *IEEE Symposium on Computational Intelligence, Cognitive Algorithms, Mind, and Brain (CCMB)*, 2011, pp. 1–7.

[11] A. Nijholt, D. Plass-Oude Bos, and B. Reuderink, "Turning shortcomings into challenges: Brain-computer interfaces for games," *Entertainment Computing*, vol. 1, no. 2, pp. 85–94, 2009.

[12] L. Farwell and E. Donchin, "Talking off the top of your head: toward a mental prosthesis utilizing event-related brain potentials," *Electroencephalography and Clinical Neurophysiology*, vol. 70, no. 6, pp. 510–523, 1988.

[13] C. Guger, S. Daban, E. Sellers, C. Holzner, G. Krausz, R. Carabalona, F. Gramatica, and G. Edlinger, "How many people are able to control a P300-based brain-computer interface (BCI)?" *Neuroscience letters*, vol. 462, no. 1, pp. 94–98, 2009.

[14] B. Allison, T. Lüth, D. Valbuena, A. Teymourian, I. Volosyak, and A. Gräser, "BCI demographics: How many (and what kinds of) people can use an SSVEP BCI?" *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 18, no. 2, pp. 107–116, April 2010.

[15] C. Guger, G. Edlinger, W. Harkam, I. Niedermayer, and G. Pfurtscheller, "How many people are able to operate an EEG-based brain-computer interface (BCI)?" *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 11, no. 2, pp. 145–147, June 2003.

[16] C. Herrmann, "Human EEG responses to 1–100 Hz flicker: resonance phenomena in visual cortex and their potential correlation to cognitive phenomena," *Experimental Brain Research*, vol. 137, no. 3, pp. 346–353, 2001.

[17] P. Allegrini, D. Menicucci, R. Bedini, L. Fronzoni, A. Gemignani, P. Grigolini, B. West, and P. Paradisi, "Spontaneous brain activity as a source of ideal 1/f noise," *Physical Review E*, vol. 80, no. 6, p. 061914, 2009.

[18] N. V. Manyakov, N. Chumerin, A. Combaz, A. Robben, and M. M. Van Hulle, "Decoding SSVEP responses using time domain classification," in *International Conference on Fuzzy Computation and 2nd International Conference on Neural Computation*, Valencia, Spain, October 2010, pp. 376–380.

[19] Y. Wang, R. Wang, X. Gao, B. Hong, and S. Gao, "A practical VEP-based brain-computer interface," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 14, no. 2, pp. 234–240, June 2006.

[20] R. Yazicioglu, T. Torfs, P. Merken, J. Penders, V. Leonov, R. Puers, B. Gyselinckx, and C. Van Hoof, "Ultra-low-power biopotential interfaces and their applications in wearable and implantable systems," *Microelectronics Journal*, vol. 40, no. 9, pp. 1313–1321, 2009.

[21] I. Volosyak, H. Cecotti, and A. Gräser, "Impact of Frequency Selection on LCD Screens for SSVEP Based Brain-Computer Interface," in *Proc. IWANN, Part I, LNCS 5517*, 2009, pp. 706–713.

[22] O. Friman, I. Volosyak, and A. Gräser, "Multiple channel detection of steady-state visual evoked potentials for brain-computer interfaces," *IEEE Transactions on Biomedical Engineering*, vol. 54, no. 4, pp. 742–750, April 2007.

[23] M. Kleiner, D. Brainard, D. Pelli, A. Ingling, R. Murray, and C. Broussard, "Whats new in psychtoolbox-3," *Perception*, vol. 36, no. ECVP Abstract Supplement, 2007.

[24] B. Rebsamen, C. Guan, H. Zhang, C. Wang, C. Teo, M. Ang, and E. Burdet, "A brain controlled wheelchair to navigate in familiar environments," *Neural Systems and Rehabilitation Engineering, IEEE Transactions on*, vol. 18, no. 6, pp. 590–598, 2010.

[25] I. Volosyak, D. Valbuena, T. Lüth, T. Malechka, and A. Gräser, "BCI Demographics II: How many (and what kinds of) people can use an SSVEP BCI," *IEEE Trans. Neural Syst. Rehabil. Eng*, vol. 19, no. 3, pp. 232–239, 2011.

[26] S. Grant, T. Aitchison, E. Henderson, J. Christie, S. Zare, J. McMurray, and H. Dargie, "A comparison of the reproducibility and the sensitivity to change of visual analogue scales, borg scales, and likert scales in normal subjects during submaximal exercise," *Chest*, vol. 116, no. 5, pp. 1208–1217, 1999.

[27] N. Lazzaro, "Why we play games: Four keys to more emotion without story," *Design*, vol. 18, pp. 1–8, 2005.

[28] P. Diez, V. Mut, E. Perona, and E. Leber, "Asynchronous BCI control using high-frequency SSVEP," *Journal of NeuroEngineering and Rehabilitation*, vol. 8, p. 39, 2011.

[29] Y. Wang, R. Wang, X. Gao, and S. Gao, "Brain-computer interface based on the high-frequency steady-state visual evoked potential," in *Neural Interface and Control, 2005. Proceedings. 2005 First International Conference on*. IEEE, 2005, pp. 37–39.

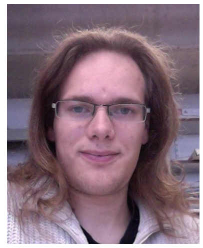
[30] F. Di Russo, W. Teder-Sälejärvi, and S. Hillyard, "Steady-state vep and attentional visual processing," *The cognitive electrophysiology of mind and brain. New York: Academic Press. p*, pp. 259–274, 2002.

[31] Y. Kim, M. Grabowecky, K. Paller, K. Muthu, and S. Suzuki, "Attention induces synchronization-based response gain in steady-state visual evoked potentials," *Nature Neuroscience*, vol. 10, no. 1, pp. 117–125, 2006.

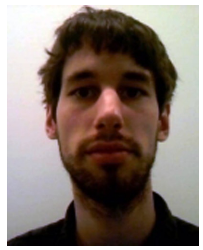
[32] I. Volosyak, H. Cecotti, and A. Gräser, "Optimal visual stimuli on LCD screens for SSVEP based brain-computer interfaces," in *Proc. 4th International IEEE EMBS Conference on Neural Engineering*, 2009, pp. 447–450.

[33] N. Chumerin, N. V. Manyakov, A. Combaz, A. Robben, M. van Vliet, and M. M. Van Hulle, "Steady State Visual Evoked Potential based Computer Gaming - The Maze," in *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering (LNICST)*, vol. 78, Genoa, Italy, May 2011, pp. 28–37.

[34] D. Regan, "An effect of stimulus colour on average steady-state potentials evoked in man," *Nature*, vol. 210, no. 5040, pp. 1056–1057, 1966.

**Marijn va Vliet** received the M.Sc. degree in human-machine interaction, which he obtained at the University of Twente, Netherlands, in 2010. During his master, he became interested in brain-computer interfaces (BCIs) and wrote his master thesis at the BCI@HMI group. He is currently a PhD student at the Computational Neuroscience Research Group at the KU Leuven (University of Leuven, Belgium), where his main work is focused on the neurological responses during language processing. His passion for BCIs however drives him to continue work on BCI implementations from time to time.



**Arne Robben** received the B.Sc. degree in Mathematics from the UHasselt (University of Hasselt, Belgium) in 2007. He completed the Mathematics program by a Master at the KU Leuven (University of Leuven, Belgium) in 2009. A year later, in 2010, he graduated in the one-year program: Master of Artificial Intelligence at the KU Leuven. Directly upon finishing his studies he joined the Computational Neuroscience Research Group at the KU Leuven. In January 2011 he obtained support from a specialization grant from IWT (Flemish Agency for Innovation through Science and Technology) to start a PhD.

**Nikolay Chumerin** received the M.Sc. degree in Mathematics and Educational Science and Higher Educational Certificate of teacher in Mathematics and Computer Science from the BrSU (Brest State University, Brest, Belarus) in 1999, and the Ph.D. degree in biomedical sciences from KU Leuven (University of Leuven, Belgium), in 2011. He is currently a post-doctoral researcher at the Laboratorium voor Neuro- en Psychofysiologie, Medical School, KU Leuven. His research interests include biologically-inspired computer vision, machine learning, signal processing and EEG-based brain-computer interfaces.

**Adrien Combaz** received the Engineer degree in Modeling and Scientific Computation from the Institut des Sciences et Techniques de lIngnieur de Lyon, Lyon, France in 2006. He then graduated from the Master of Artificial Intelligence from the KU Leuven (University of Leuven, Belgium) in 2008. He is currently a doctoral student at the Computational Neuroscience Research Group, Laboratorium voor Neuro- en Psychofysiologie, Medical School, KU Leuven. His research focuses on neuroscience, machine learning, signal processing and EEG-based brain-computer interfaces.

**Nikolay Manyakov** received the M.Sc. degree in mathematics from the BSU (Belarusian State University, Minsk, Belarus) in 1998, the Ph.D. degree in theoretical informatics from the BSUIR (Belarusian State University of Informatics and Radioelectronics, Minsk, Belarus) in 2005, and the Ph.D. degree in biomedical sciences from KU Leuven (University of Leuven, Belgium), in 2010. He is a Research Fellow at the Computational Neuroscience Research Group, Laboratorium voor Neuro- en Psychofysiologie, Medical School, KU Leuven, Belgium. His current research interests include brain-computer interfaces, computational neuroscience, neural networks, machine learning, data mining, and signal processing.

**Marc M. Van Hulle** received the M.Sc. degree in electrotechnical engineering and the Ph.D. degree in applied sciences from the KU Leuven (University of Leuven, Belgium). He also received the B.Sc.Econ. and M.B.A. degrees. He received the Doctor Technices degree from Queen Margrethe II of Denmark, in 2003, and an Honorary Doctoral degree from BrSU (Brest State University, Brest, Belarus), in 2009. He is currently a Full Professor at the Medical School of the KU Leuven, where he heads the Computational Neuroscience Research Group of the Laboratorium voor Neuro- en Psychofysiologie.